Information Technology University



Lab Manual

for

CS220: Artificial Intelligence

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Semester Fall 2023 : **3**
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Task 1: Visualizing Data

In this task, you will use **Matplotlib**, a Python library for creating static, animated, and interactive visualizations. You can import it as following:

```
import matplotlib.pyplot as plt
```

a) Display a Single Random Image

In this part, you will display one random image from the training dataset with its label as the title.

Steps to Visualize a Single Image:

- **Pick a Random Image:** Select an image and its corresponding label randomly from the training dataset.
- **Plot the Image:** Use `plt.imshow()` to display the image.
- **Add a Title:** Use `plt.title(label)` to add the label as the title of the image.
- **Show the Image:** Call `plt.axis('off')` to remove axes, and `plt.show()` to display the image.

b) Visualize Data

Define a function `visualize_data(train_images, train_labels, rows, cols)` to visualize MNIST images.

Steps to Visualize Data:

1. **Create a Grid of Subplots**
   Use `plt.subplots(rows, cols)` to prepare a grid for displaying multiple images.
2. **Randomly Select Images**
   Randomly pick images and their corresponding labels from the dataset.
3. **Display Each Image**
   Use `ax.imshow()` to display each image in grayscale and `ax.set_title()` to add the label as the title.
4. **Hide Axes**
   Call `ax.axis('off')` to remove unnecessary ticks and labels for a cleaner look.
5. **Show the Plot**
   Use `plt.tight_layout()` to adjust spacing and `plt.show()` to display the grid without overlaps.

Task 2: Counting and Visualizing Data Distribution

In this task, you will analyze the distribution of data in the MNIST dataset by counting how many images belong to each label.

a) Count the Number of Images per Label

In this part, you will count how many images exist for each label (from 0 to 9) in the dataset.

Steps to Count Images per Label:

1. **Initialize a Counter:** Create an array or list to count occurrences of each label.
2. **Loop Through Labels:** Iterate through the `train_labels` and update the counter for each label.
3. **Display the Counts:** Print the number of images for each label from 0 to 9.

b) Create a Histogram of the Label Counts

In this part, you will create a histogram to visualize how many images belong to each label.

1. **Steps to Create the Histogram**:.
    - **Plot the Histogram:** Use `plt.bar()` to plot the counts of each label.
    - **Add Labels:** Set the x-axis to represent the labels (0-9) and the y-axis to represent the count.
    - **Display the Plot:** Add a title and labels, then call `plt.show()` to display the histogram.